

Project Planning Phase

Project Planning Template

Phase 3: Project Planning Phase

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Agile Development Methodology

Development followed the Scrum framework, with 2-week sprint iterations and a planned team velocity of 10 Story Points per sprint. All user stories are traced from the Requirement Analysis phase.

2. Product Backlog and Sprint Schedule

Sprint	Epic	USN ID	User Story / Task	Story Points	Priority	Status
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Sprint 1	Foundation and Auth	USN-001	Project setup: Express API server, React frontend structure, MongoDB initial model bindings.	3	High	Completed
Sprint 1	Foundation and Auth	USN-002	Secure user registration with Bcrypt hashing and validation checks.	2	High	Completed
Sprint 1	Foundation and Auth	USN-003	User login session initialization returning signed JWT tokens in HTTP-Only cookies.	2	High	Completed
Sprint 2	Discovery and Cart	USN-004	Product catalog search with live autocomplete suggestions and category filters.	3	High	Completed

Sprint 2	Discovery and Cart	USN-005	Persistent shopping cart context preserving products, totals, and quantities on page reload.	3	High	Completed
Sprint 2	Discovery and Cart	USN-006	Product wishlist management:: saving favorites and supporting One-Click Move to Cart.	2	Medium	Completed
Sprint 2	Checkout and Fulfill	USN-007	Multi-step checkout wizard integrating active coupon rules validation and tax estimations.	3	High	Completed
Sprint 3	Checkout and Fulfill	USN-008	Stripe mock integration to generate payment intents and verify checkouts.	3	High	Completed

Sprint 3	Feedback and Logs	USN-009	Order history lookup and visual timeline fulfillment tracking.	2	Medium	Completed
Sprint 3	Feedback and Logs	USN-010	Verified buyer reviews and helpful feedback upvote system.	2	Medium	Completed
Sprint 3	Administration	USN-011	Unified Admin Dashboard panel with sales charts, inventory tables, and coupon controls.	5	High	Completed

3. Detailed User Stories and Acceptance Criteria

USN-002: User Registration

As a new customer, I can register a profile by entering my name, email, and password, so that I can shop and track orders.

Acceptance Criteria:

- Must block submissions with duplicate emails or weak passwords (less than 6 characters).
- Password must be securely encrypted using Bcrypt before saving to MongoDB.
- Successful registration redirects the user directly to the home screen with an active session.

USN-004: Autocomplete Product Search

As a shopper, I can input search words and select categories, so that I can find matching products immediately.

Acceptance Criteria:

- Submitting text queries instantly queries the /api/products/suggestions API.
- Results must respect category selections, price ranges, and minimum rating star filters.
- Sorting options (price, rating, date) must re-order results immediately on-screen.

USN-007: Coupon Code Validation

As a customer, I can apply coupon codes during checkout, so that I receive active promotional discounts on my orders.

Acceptance Criteria:

- Coupon validation endpoint /api/coupons/validate checks active flags and expiry dates.
- Applied discounts must be itemized (fixed vs percentage) and deducted from the subtotal.
- Taxes (15% VAT) and shipping speeds (Express vs Standard) must recalculate automatically.

USN-011: Unified Admin Dashboard

As a store admin, I can view visual monthly sales graphs and inventory tables, so that I can manage my shop operations.

Acceptance Criteria:

- Metrics charts must compile monthly revenue, category shares, and new user enrollments.
- Product inventory tables must render alerts for items below stock thresholds.
- Admin can instantly block/unblock customer logins and toggle coupon statuses.

4. Sprint Metrics and Velocity Tracker

Sprint	Duration	Planned SP	Completed SP	Focus Areas
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Sprint 1	Weeks 1-2	7	7	Base Express structure, database models, user registry, JWT sessions
Sprint 2	Weeks 3-4	11	11	Product catalogs, search suggestions, persistent cart, address books, coupon validation
Sprint 3	Weeks 5-6	12	12	Stripe mock payments, visual order tracking, buyer reviews, Admin analytics dashboard

Average Velocity Achieved: 10 Story Points per Sprint. Development Methodology: Scrum framework with 2-week sprint iterations.